From Static Prompts to Agentic Ranges: A Taxonomy of LLM Cybersecurity Benchmarks

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This paper focuses on the taxonomy of Large Language Models (LLM) cybersecurity benchmarks and how their results should not be used as a single measure of capability. The focus of this approach is to understand how the design choices of the benchmarks shape the results and the meaning of the results.

This paper includes a literature survey of large language models and LLM-based agents and their cybersecurity capabilities. This paper focuses on cyber-specific benchmarks that evaluate LLMs or LLM-based agents across examples such as “capture the flag” tasks, prompt-injection refusal behavior, and cybersecurity exam-style questions (Bhatt et al., 2024; Tihanyi et al., 2024; Zhang et al., 2024). The papers were chosen based on their specific focus on what the benchmark is measuring. The paper’s purpose is not to introduce new original findings or extend the work from past benchmarks but rather synthesize the taxonomy of benchmark measurement choices from 2021 to 2026. By organizing benchmarks around taxonomy, cyber benchmarks can be evaluated based on how the benchmark measures LLMs' or LLM-based agents' performance.

The papers that were included fall under specific requirements. Papers that counted were required to offer a specific contribution in the AI field that establishes explicit benchmarks or scores for the LLMs or LLM-based agents. The research must also be specific to LLMs and how they perform on cybersecurity tasks. General performance benchmarks that include cybersecurity were not included. By not including general performance papers the taxonomy of the selected papers can be evaluated in an exclusive way. Benchmark papers used were limited to those published on arXiv for consistency of sources and to ensure availability.

This paper organizes the corpus around the taxonomy and design of each paper. This includes the paper covering insecure code generation, “Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions” by Pearce et al., and the CyberMetric Benchmark (Pearce et al., 2021; Tihanyi et al., 2024). CyberSecEval 2 and 3 cover prompt, tool, and general safety (Bhatt et al., 2024; Wan et al., 2024). The next three papers, Cybench, CVE-Bench, and Folkerts, contribute findings on the agentic/offensive and long-horizon work (Folkerts et al., 2026; Zhang et al., 2024; Zhu et al., 2025). CyberSOCEval focuses on defensive malware analysis and threat-intelligence reasoning, and CTFusion evaluates live CTF challenges performed by agents (Deason et al., 2025; Lee et al., 2026).

The corpus should not be treated as a complete cybersecurity survey; the corpus itself is limited to some of the major LLM-based cybersecurity benchmark issues. It was chosen as a base for surveying taxonomy within cyber benchmarks as they evolve, not as a chronological timeline. A strong agentic cluster is observed in this paper, but not all works cover this. Social engineering is present but is not covered deeply. The cutoff date of June 2026 is noted so further advancements or benchmarks were not included.

Taxonomy Design and Dimension Overlap

This survey follows a concept-centric strategy by arranging the work around dimensions instead of the publication date of the paper (Webster & Watson, 2002). Timelines often overemphasize evolution with a cause-and-effect relationship within a timeline. By using taxonomy, the survey can compare approaches in a scientific way rather than a historical mindset. This choice makes it easier to compare, construct, and test the differences in each paper and its benchmark. This helps the reader understand the progression of benchmarks is not linear

over time but rather each benchmark can be evaluated specifically on its design choices and limitations.

The taxonomy used in this paper has three dimensions. The first dimension is construct plus role, which is about what the benchmark is claiming to measure. Second, the elicitation system is about how a benchmark is producing the model's behavior. Third, validity controls focus on whether the measurement of the model can be trusted. Together, these dimensions break down the evaluation of these papers to be compared by their design and not their individual results. This also avoids the trap of only using a chronological approach by publication date and comparing them on a timeline. The heatmap in Figure 1 highlights how coverage and taxonomy were used to evaluate the benchmarks. It provides evidence of how the benchmarks were scored, not on their score, results, or chronological basis.

Figure 1

Coverage Matrix Across the Selected LLM Cybersecurity Benchmark Corpus

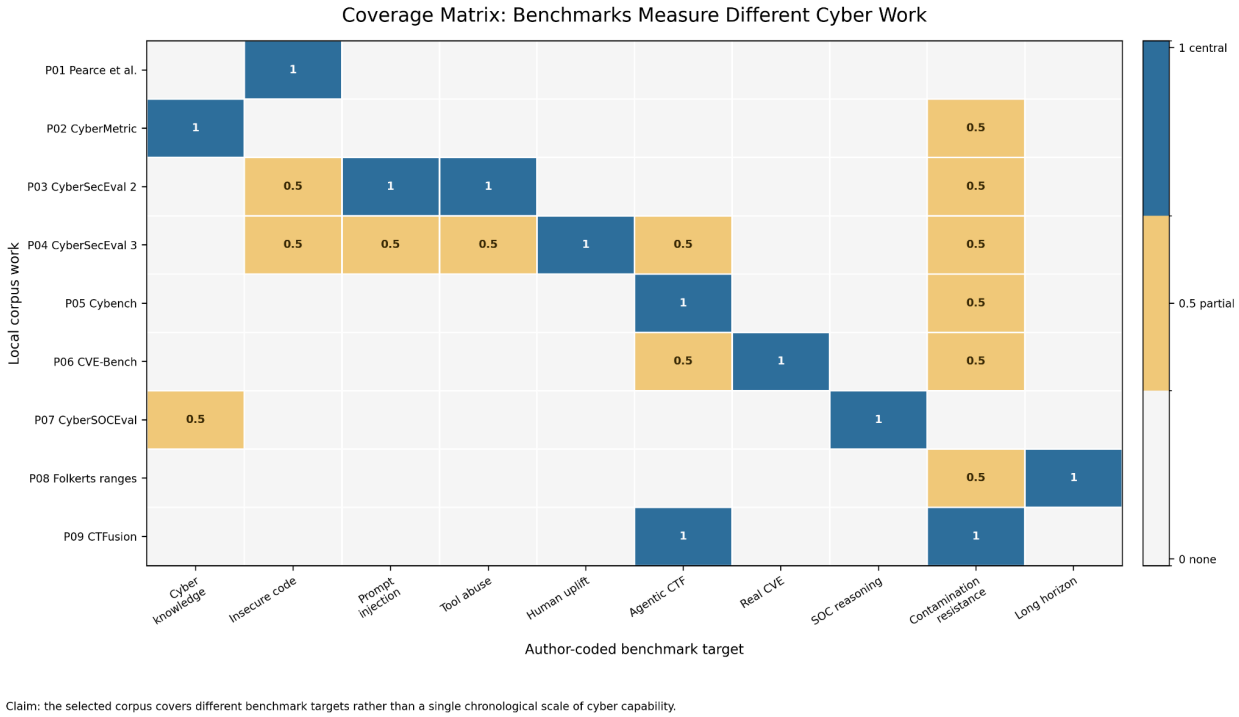


Figure 1 shows the breakdown of how each benchmark paper was scored using a coverage matrix. Papers with a 1 qualified under being the main focus of the benchmark. The main focus is the cyber capability being tested in the benchmark. 0.5 describes a partial coverage where it partly covers an area that the benchmark tests. The score of 0 means it was not covered on the benchmark. The choice of three dimensions was a decision to develop a synthesis for cybersecurity benchmarks, while also accounting for limitations in scope. By narrowing down to three dimensions, the framework is separated enough to break down the benchmarks and their implicit design choices. The dimensions are encompassing enough to include the benchmarks in this survey, but exclusive enough to block general benchmarks that include a cybersecurity component. It keeps the corpus of the survey in the correct space, narrow enough to avoid drifting out of scope. The different dimensions are intended to limit overlap in the categories that they cover. This is designed to keep each section from becoming too repetitive.

The context given and the environmental sandbox employed by the benchmarks create a regime-dependency that does not always clearly map across all the included papers. Low-tool benchmarks like CyberMetric, CyberSOCEval, and Pearce limit the additional context given to the LLM, like web tools or live web applications (Deason et al., 2025; Pearce et al., 2021; Tihanyi et al., 2024). Benchmarks like Cybench, CVE-Bench, Folkerts, and CTFusion use agentic and high-tool designs (Folkerts et al., 2026; Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). These design choices show taxonomy is not always unique and benchmarks often draw upon their predecessors. It also explains how results could be regime-dependent in these benchmarks. These results refer to the benchmark results for each paper. Comparison across regimes highlights that taxonomy is not unique, and the dimensions are not always independent.

The taxonomy is useful because of how benchmarks overlap in dimensions. Benchmarks with tool usage have similar elicitation systems with different constructs. This would cause repetitive arguments across the body of the paper since environments often share a similar elicitation system. In addition, since dimensions are not always independent as collinearity shows, taxonomy fulfills the context needed that a chronological timeline would lack. This develops a clear organization around a nonrepetitive and contextual approach of taxonomy.

Construct Plus Role: What Cyber Work Is Operationalized

The first taxonomy dimension is construct plus role, because benchmarks vary in what cybersecurity work they ask the model to perform. In Pearce et al., the benchmark constructs model output as code suggestions inside VSCode using GitHub Copilot. Copilot acts as an AI coding assistant focusing on suggesting code to the user. The authors inquired about the cybersecurity implications of auto-suggested code based on the role given to Copilot of an AI coding assistant. The benchmark's output specifically tracked the percentage of suggested code that was vulnerable to cyber exploits. It found that Copilot offers vulnerable code about 40% of the time (Pearce et al., 2021). This result shows a sizable gap between the role given to Copilot and its output. Since the model only suggested code and did not have agentic capabilities, this low-tool benchmark provides an example of a benchmark that tests real-world suggestions in a low-tool way. Comparing the prior benchmark to CyberMetric provides the contrast in taxonomy between the low-tool-use benchmarks. CyberMetric provides a benchmark output of overall cybersecurity knowledge based on a multiple-choice exam format (Tihanyi et al., 2024). This is low-tool use because the benchmark is only testing general knowledge in an academic and conceptual way. The questions included in the benchmark are theoretical in nature, and are not real systems the model can interact with in an environment. Although limited in scope, the

low-tool benchmarks still measure different dimensions of LLM cybersecurity ability with Pearce et al. focusing on insecure code generation as an AI coding assistant and CyberMetric as a multiple-choice exam covering general cybersecurity knowledge of an LLM.

The next change is the role shift of benchmarks testing the output or knowledge of a model, rather than focusing on whether the model can police itself and recognize attacks from users. In CyberSecEval 2, the benchmark gives a model the role of protecting itself from prompt injection, code-interpreter abuse, false refusal, etc. The benchmark uses 500 interpreter-abuse prompts to test the model's internal safeguards (Bhatt et al., 2024). This contrasts the previous benchmarks' assigned roles and tasks for the LLM. This benchmark's output shows whether the model can safely police itself given malicious inputs. CyberSecEval 3 expands on the second version by increasing the scope of the testing to eight risk areas (Wan et al., 2024). These new additions include spear phishing, manual uplift, and autonomous cyber operations. While both versions test for safety failures and attempt to understand what the model will generate when faced with malicious inputs, they differ in scope and completeness. As the risk areas quickly evolve in AI cybersecurity, the demand for new versions is clear given the organization and strategy behind these two renditions of the benchmark. The benchmarks also provide examples of the change in role between testing the LLM output and testing whether the LLM will behave safely.

The next change in benchmarks places the model as an autonomous offensive agent. The benchmarks Cybench, CVE-Bench, and Folkerts all place the model as an offensive agent given the role to accomplish a task. In Cybench, the agent's role is to complete 40 professional capture-the-flag (CTF) challenges (Zhang et al., 2024). CVE-Bench tests the agent's ability in 40 web CVEs (Zhu et al., 2025). Lastly, Folkerts offers two cyber ranges, one with 32 steps and the

other 7 steps (Folkerts et al., 2026). While testing offensive capabilities, the target of the benchmarks changes from a CTF task, to web applications, to long-horizon range tasks.

CyberSOCEval gives the model the role of defensive SOC. This reasoning role instead of an offensive agent constructs a different target forcing the model to reason and provide accurate defensive cybersecurity results. This benchmark's goal is to accurately output a result that tests a model's ability to reason in a defensive scenario. It offers 609 malware-analysis validated cases (Deason et al., 2025). CTFusion shows live governance results. Folkerts' design to evaluate longevity concerns also tests the autonomous ability of models and how they reason over a long task period (Folkerts et al., 2026). These roles extend beyond a coding assistant or offensive agent roles because the benchmarks specifically output results that test governance, not pure output.

Elicitation System: Capability, Scaffold, Tools, Environment

The elicitation systems across the benchmarks shift from answering questions and writing code to the ability to act inside an environment. When a benchmark goes from a test of general cybersecurity knowledge to inserting a model into a task environment, the output of the benchmark changes. In Cybench, CVE-Bench, Folkerts, and CTFusion, the model is being evaluated within environmental constraints (Folkerts et al., 2026; Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). When reading the results, the environment, tools, and scaffold are necessary to account for. Compare this to when testing in a multiple-choice setting like CyberMetric. The LLM does not have tool access to the web to find answers (Tihanyi et al., 2024). The benchmark is only drawing answers out of the model's knowledge, not from its tool use. This divide in elicitation systems shows the change in how benchmarks approach producing outcomes and how they measure the models.

The restriction in agency within the benchmark is a qualifying characteristic that shows the differentiation of the elicitation systems across benchmarks. In CyberMetric the use of a multiple choice exam rather than an interactive environment (Tihanyi et al., 2024). The multiple-choice exam focused on depth of knowledge which is useful, but failed to accurately account for the model's ability to chain together multiple actions to produce an outcome. In Pearce, the coding agent can offer suggestions but cannot test the suggested code, leaving a gap between secure code and suggested code from Copilot (Pearce et al., 2021). The static output of these benchmarks contributes useful information regarding LLM cybersecurity ability as a reference point for further testing. The limitations of the two benchmarks are that they measure static output rather than what the model can handle interactively.

In Cybench, CVE-Bench, Folkerts, and CTFusion, models are given an environment with the agency to interact within it (Folkerts et al., 2026; Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). The access to a terminal, tool use, and how the challenge is presented to the model all impact the measured object. For example, in CTFusion, the use of an MCP gives it access to live challenges and artifacts, while an additional system was used to validate the flags (Lee et al., 2026). A model could score high on a static output benchmark, but struggle using MCP tools, thus weakening the model's interaction channel. A model's behavior between challenges also differs. In Folkerts the long-sequence executions in extended cyber ranges highlight the tool usage in the benchmark. These benchmarks show what a model can do in an environment over time, not just that it has the ability to produce the right answer when asked.

Once the success of the benchmark depends on the model and scaffolding together these agentic benchmarks transition to measuring system-level performance rather than base-model performance alone (Zhang et al., 2024; Zhu et al., 2025). In CVE-Bench, the introduction of web

applications within the benchmark creates an example of a measurement artifact (Zhu et al., 2025). These skills along with the benchmark's results produce a distorted view of how successful the scaffold is at getting the model to produce a successful outcome. This artifact is less visible in benchmarks producing static output and is a result of benchmarks shifting to approaches using environments. This evolves these benchmarks with agentic structure into a systems performance test, not just a base-model benchmark. This growth supports that elicitation design has changed over benchmarks and their output has become deeper in contextual design.

Not all benchmarks fall into these two categories; some fall somewhere in the middle. CyberSecEval 2 and CyberSecEval 3 are benchmarks that are in this middle category. CyberSecEval 2 shifts from Pearce, which focuses on suggesting vulnerable code, to CyberSecEval 2 which focuses on how well a model can follow a prompt (Bhatt et al., 2024; Pearce et al., 2021). Because CyberSecEval 2 centers on prompt injection, it tests the model's ability to follow directions or safeguard itself against malicious requests. This shows the benchmark’s elicitation design is to create output that gives context around how the model performs without an environment to actually test what it can do. CyberSecEval 3 involves both static evaluations and autonomous tool use through cyber operations assessments (Wan et al., 2024). Model outputs from these benchmarks do not have the degree of measurement artifact as high-tool environment benchmarks. This creates a connection between low- and high-tool-use benchmarks showing that elicitation design is a spectrum and no one-size-fits-all solution exists.

The takeaway from this section is the elicitation design changes the meaning of the benchmark's results. A multiple-choice exam and an environment with web applications do not test the same kind of “cyber capabilities” and do not produce the same results (Tihanyi et al., 2024; Zhu et al., 2025). These results cannot be directly compared broadly with other

benchmarks without understanding what the benchmark is actually measuring. This is why elicitation design is one of the three dimensions of taxonomy.

Validity Threats and Controls

The dimension of validity matters because it answers how benchmark scores can be the result of benchmark design choices and not the model's cybersecurity capability. Just because a benchmark produces a score does not mean that score can be assumed to be the full cybersecurity capability of a model. Rather, the proxy, judgment methods, scaffold effects, and sandbox realism have to be checked to ensure the validity of the benchmark. By validating the benchmark environment, the validity of the score and the meaning of the benchmark results can actually be trusted.

With different benchmarks employing different proxies for cyber capability, those proxies are limited in scope if applied as an encompassing cyber benchmark of a model's capabilities. Are the multiple-choice exams given in CyberMetric, CTF challenges, and CVE exploitation all evaluating the same thing? The result is that an exam and a sandboxed benchmark with challenges test different parts of a model's cybersecurity capability (Tihanyi et al., 2024; Zhang et al., 2024; Zhu et al., 2025). Each benchmark focuses on its own aspects for its cyber capability with its output’s validity tuned to its design. This shows that not a single benchmark covers all cyber capabilities, and a multi-benchmark approach offers the most nuance in establishing this.

Contamination is a critical component in determining the validity of a model's benchmark results. Benchmarks like CVE-Bench and CTFusion focus efforts to ensure their benchmark tasks are not leaked into training data or writeups (Lee et al., 2026; Zhu et al., 2025). This keeps the benchmark's validity for contamination as intact as possible. Cybench is a benchmark that even acknowledges it is hard to eliminate contamination in all models, showing how serious of

an issue contamination is to the benchmark's validity (Zhang et al., 2024). This shows the importance of acknowledging contamination as a threat and the efforts taken to minimize contamination within the benchmark.

The next aspect of validity is who decides whether the model succeeded during the benchmark. Benchmarks like CyberSecEval 2 and 3 assign LLM judges to help assist in scoring (Bhatt et al., 2024; Wan et al., 2024), but CyberSecEval 3 also validates its spear-phishing judge against smaller blind human evaluation (Wan et al., 2024). Other benchmarks use human validation, while others in environments with tools and applications judge the success of the model based on partial progress (Deason et al., 2025; Zhang et al., 2024; Zhu et al., 2025). With each benchmark using different judges and metrics for success, the validity of each benchmark cannot be assumed across the category for being broadly cyber and must be verified as to which cyber proxy it addresses.

When benchmarks include scaffolding to complete tasks, the benchmark score is based on the scaffolding and the model, not just the model itself. More scaffolding offered to the agent can also increase the scaffolding confounding in the benchmark. In Cybench, the scaffolding offered to models helped some and hurt others, this result calls into question whether more scaffolding is always beneficial for a model (Zhang et al., 2024). In CVE-Bench, teams of agents and AutoGPT outperformed Cy-Agent during testing. This is another example of how scaffolding variants can change the outcome of the model's performance (Zhu et al., 2025). Testing different scaffolds within the same benchmark gives the opportunity to check the validity of the model's scores without solely relying on one scaffold and the model's output.

Although benchmarks strive to test the full extent of a model's cyber-specific capabilities, the scores should be interpreted as a model's ability inside that benchmark environment. For CTF

challenges, these tasks are time-bounded and contain smaller codebases than real-world scenarios would include (Zhang et al., 2024). Benchmarks even state that their challenges are designed specifically to be buildable and solvable (Zhang et al., 2024). These factors point to taking nuance in reading cyberbenchmarks and to avoid broad assumptions about a model's cyber strength based on just one benchmark. Even more recent and encompassing benchmarks have limitations, with tasks limited to popular exploits (Zhu et al., 2025). Understanding the context and limitations of a benchmark helps establish its validity relative to other benchmarks in the cybersecurity space.

Findings

The selected corpus shows the expansion over all three taxonomy dimensions from static output to expanded benchmarks testing how models complete cyber-related tasks. Pearce only suggested code, and CyberMetric used exam answers to produce the output (Pearce et al., 2021; Tihanyi et al., 2024). The benchmark score was specific to the model and what was covered in the benchmark. This expands to benchmarks with an elicitation system that produces an output that tests the model's ability within a scaffold to complete cyber tasks such as in Cybench, CVE-Bench and CTFusion (Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). Not all benchmarks fall into these two categories with CyberSecEval 2 and 3 offering a mixture of some scaffolding while keeping the results more model specific (Bhatt et al., 2024; Wan et al., 2024). With the different designs in what the benchmark's output produces, validity controls must be scrutinized and adapted on a benchmark-by-benchmark basis analyzing what the benchmark's output means and recognizing it doesn’t account for real-world performance.

The second finding is that validity increasingly depends on contamination and retrieval leakage across benchmarks (Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). The common

trap of publicly released benchmarks is that once they are public, they enter the realm of information that models can train on (Lee et al., 2026; Zhang et al., 2024; Zhu et al., 2025). This drives the demand for new benchmarks because of the threat of contamination. Including web search and other tools increases the possibility for leakage and contamination, creating challenges for new benchmarks. This contrast in how low-tool and high-tool benchmarks handle contamination is central to evaluating benchmarks across the corpus.

Limitations

Limitations of the survey are that this corpus is limited in size and more cybersecurity benchmarks exist that were not included in the paper. Another limitation is that benchmarks chosen were specifically selected. CyberMetric being the only knowledge anchor is also a limitation of the corpus. The results of this survey are not statistically representative of the broader cybersecurity ability.

Conclusion

The corpus of benchmarks can best be understood using a three-dimensional taxonomy approach. By avoiding the pitfalls of a chronological timeline, the paper uses elicitation system design, the way cyber work is organized, and how validity is protected to classify benchmark results. To answer the paper's question this shows that cybersecurity benchmarks for LLMs cannot be compared across scores, and must be understood as a single score for the cyber area it is testing. By better understanding benchmarks and their results, cybersecurity abilities can be better understood and recognized and avoid validity issues when comparing across benchmarks solely using the benchmark scores. Benchmarks are not a solution to show a model's real-world capabilities, but show what a model's ability is in that specific cyber area. As benchmark

scaffolding and tasks expand in scope, questions of validity and controlling leakage will become more important to ensure the benchmark's results are usable (Lee et al., 2026; Zhu et al., 2025).

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